ADMINISTRATOR MANUAL

DialerOS

Fully-local call-center platform — telephony, outbound/inbound, agents, and the Master/Worker AI agent.

Operations & configuration guide for administrators & supervisors.
Covers v1.x & the Phase L AI agent (iters 199–215).
Host: dialeros.voipzap.com · All inference is local (no external AI/API).

Contents

1. What DialerOS is & how it fits together

DialerOS is a self-hosted, **fully-local** call-center platform: predictive/progressive outbound dialing, inbound ACD queues, IVR, agents (human *and* AI), recording, QA, compliance, and a multi-node cluster — with an AI agent stack that runs entirely on your own hardware (no external AI service is ever called).

The moving parts

Component	Role
`admin-gui` (this web app, port 1111 behind nginx)	Everything you configure & monitor. Next.js app.
`control-plane`	The business logic + SQLite data store the app uses.
`FreeSWITCH`	The telephony engine: places/receives calls, bridges audio, plays IVR.
Nodes	One or more telephony hosts. DialerOS is cluster-aware (load is balanced + visible).
Local AI stack	Ollama (LLM), whisper.cpp (speech→text), Coqui XTTS (text→speech), all-minilm (embeddings), and a media-bridge daemon. **All on your box(es).**

> **Golden rule:** nothing leaves your infrastructure. Every AI inference (speech, language, embeddings) is local. The only "external" traffic is your SIP carriers carrying the actual phone calls.

2. Logging in & roles

Open `https://dialeros.voipzap.com` and sign in. First-ever boot uses the `/setup` page to create the initial admin.

Roles & user levels

Role	Can do
admin	Everything: config, users, AI, settings, cluster.
supervisor	Monitor live calls, take over/seize, callbacks, reports (read-only config).
agent	Take/place calls on the agent screen only.
operator	Back-office; no telephony.

Fine-grained permissions (the ACL matrix) and a numeric **user level (1–9, ViciDial-style)** are set per user on their detail page. A ticked permission stays inert until the user's level meets its minimum — this lets you template access by level. Admins implicitly have everything at level 9.

3. Monitoring: Dashboard, Realtime, Supervisor

Dashboard (`/`)

Landing overview — system health, today's call volume, active campaigns/agents at a glance. Your first stop each shift.

Realtime (`/realtime`)

Live pulse of the floor: calls in progress, agents by state (available / on-call / wrap-up / paused), queue depth and wait, pacing pressure. Use it to spot a stalled campaign or a queue backing up.

Supervisor (`/supervisor`, `/supervisor/calls`, `/supervisor/callbacks`)

ViciDial-style monitor. See paused-or-active agents and live calls; **listen (eavesdrop)**, **whisper/coach**, **barge**, or **seize** a call — including taking over an AI-handled call (the AI hands the live call to you). Manage queued callbacks here too.

> Supervisor monitoring uses the browser softphone (WebRTC) — no extra desk phone needed.

4. Telephony foundation

Set these up **in this order** before any dialing.

Carriers / trunks (`/carriers`)

Your SIP providers. Add each carrier (gateway host, auth, channel limits, dial-string rules). DialerOS supports **carrier racing** (fork to multiple carriers, first to answer wins) with auto-prune of flaky carriers (see Settings → Carrier-race auto-prune).

DIDs (`/dids`)

Inbound phone numbers you own. Point each DID at an in-group (queue), an IVR/call-menu, or a campaign.

CID groups (`/cid-groups`)

Pools of outbound caller-ID numbers. Rotate CIDs per campaign for local-presence dialing and to spread call volume (works with frequency caps for compliance).

Route plans (`/route-plans`)

How an outbound call chooses a carrier + CID: ordered rules, failover, parallel/race options, area-code/destination matching. Campaigns reference a route plan.

5. Inbound

In-groups (`/in-groups`)

Inbound ACD queues. Each in-group has DIDs, a routing strategy, business hours/timezone, priority, queue announcements, and assigned agents (human *or* an AI-agent user — see §9). Callers route here; the longest-waiting/priority caller is matched to the next free agent.

Call menus / IVR (`/call-menus`)

DTMF menus ("press 1 for sales"). Build a tree of prompts → actions (route to in-group, play audio, voicemail, hang up). Attach a call-menu to a DID for an auto-attendant.

Callbacks (`/callbacks`, Settings → Callback)

"Don't wait on hold — we'll call you back." Callers can request a callback via DTMF; the request is queued and dialed back when an agent is free. Configure the offer/DTMF in Settings → Callback.

6. Outbound

Campaigns (`/campaigns`)

The outbound engine. Create a campaign, then configure:

- **Dial mode & pacing** — predictive/progressive ratio, abandon target (global defaults in Settings → Pacing).
- **Route plan + CID group** — how it dials out.
- **Dispositions** (`/campaigns/[id]/dispositions`) — call outcomes agents pick; drive recycle/DNC logic.
- **Survey** (`/campaigns/[id]/survey`) — scripted survey + responses report.
- **AI persona binding** — optionally bind a Worker-AI persona (and an A/B challenger) so answered calls go to the AI instead of a human bridge (see §9).

Leads (`/leads`)

Import/manage the calling list. Add leads, see per-lead history/dispositions, recycle rules. Bulk import on `/leads/add`.

DNC (`/dnc`)

Do-Not-Call list. Numbers here are never dialed. Dispositions can auto-add to DNC.

Compliance — TCPA, consent, frequency caps

Settings → Frequency caps (max attempts per lead/CID per window), Consent records (`/consent-records`), TCPA report (`/reports/tcpa`), Holidays (no-call dates). Configure these before going live in regulated markets.

7. Agents

Users (`/users`, `/users/add`, `/users/[id]`)

Create human or **AI-agent** users. On a user's detail page:

- **Access** panel — role, user level (1–9), ACL matrix.
- **AI agent** panel (admin-only) — set *Account type = AI agent* and pick a *persona*. This is how a Worker AI becomes an "agent" you can assign to in-groups/campaigns exactly like a person. (Added iter 210 — also settable at user creation.)
- **Phones** — SIP credentials; primary phone is what the softphone registers / the pacer bridges to.
- **Skills / In-groups / Campaigns** — what the user is attached to.

Remote agents (`/remote-agents`)

Agents that answer on an external number/SIP rather than the browser softphone (work-from-home via PSTN). Capacity-tracked like local agents.

The agent screen (`/agent`)

What a human agent uses on shift: softphone, screen-pop, disposition, wrap-up. Wrap-up enforcement is configurable (Settings → Wrap-up enforcement).

8. Audio

Audio Center (`/audio-center`)

Cluster-wide library of every audio file (prompts, MOH, voicemail drops) with details + a usage cross-reference (where each file is used). Upload here; files are normalized to FreeSWITCH-friendly format automatically.

Sound board (`/sound-board`)

One-click soundbites an agent can play into a live call (legal disclosures, hold message, etc.).

9. The AI agent (Master / Worker)

This is DialerOS's headline capability: an AI that **takes real calls assigned exactly like a human agent**, fully local, and **inert until you deliberately arm it**.

Concepts

Term	Meaning
Worker AI	The thing on a call: speech→text (whisper) → LLM (Ollama) → text→speech (Coqui), guarded + scrubbed.
Master AI	The global brain: scoped memory (RAG), auto-curated exemplars from great calls, learned transfer rules, QA — shared across Workers.
Persona	The agent's name, designation, script, voice, model, guardrails. Managed on `/settings/ai-personas`.
Identity discipline	The agent states *only* the name/designation you give it and **never reveals it is an AI** — enforced by a hard prompt guard *and* a reply scrub.

AI personas (`/settings/ai-personas`)

Create/edit a persona: *Name*, *Agent name it states*, *Designation*, system prompt/script, greeting, LLM model, STT model, TTS engine/voice, max turns, max call seconds, escalation keywords. **Use the per-row Enable/Disable toggle** to make it live (the backend always supported this; the toggle was added iter 210). The built-in *Sandbox* lets you text-test a persona instantly.

Master AI page (`/settings/ai-master`) — the control center

1. **Master enable** — coordination scaffolding switch.
2. **AI readiness preflight** — a live checklist of every dependency (Ollama + models, whisper, Coqui, FreeSWITCH `mod_audio_stream`, a bound persona, the live switch) with the exact remediation per item, and an *armed / not-armed* verdict. **Always check this first.**
3. **LLM provider** — choose the local inference engine: Ollama (default) or any local OpenAI-compatible server (llama.cpp-server / vLLM). **This is the one-field scaling lever** — point Base URL at a stronger node and everything (calls, mock, training) gets faster/better with no redeploy. A non-local URL is rejected by design.
4. **Master memory (RAG)** — knowledge the agent retrieves at call time, scoped global / campaign / in-group. Enable/disable/delete entries.

5. **Training Center** — teach the agent four ways: type knowledge, upload an audio recording (whisper-transcribed), ingest a real ended call, or let the Master interview you. Everything is embedded locally and used on the next call.
6. **Mock Call** — test the agent as a customer, in *text* or *live hands-free audio* (it greets you, you talk, it replies in voice, loops). Uses the real pipeline; nothing is saved.
7. **AI performance** — latency knobs: reply length, creativity, keep-warm, prompt budget, TTS speed.

How to put an AI agent on the phones

1. Create + **enable** a persona (name, designation, script, voice). Test it in Mock Call.
2. Create an **AI-agent user** (Users → Add → Account type = AI agent → pick the persona), or set it on an existing user's *AI agent* panel.
3. Assign that AI user to an **in-group** (inbound) and/or bind the persona to a **campaign** (outbound) — exactly like a human.
4. Clear every blocker on the **readiness preflight** (esp. `mod_audio_stream` — a one-time operator step: `scripts/install-audio-fork.sh`).
5. Flip **ai.live_enabled** (the master live switch). Until this is on, the AI is completely inert.

> By design the AI is the **overflow** on inbound: a free human always wins; the AI takes the caller only when no human is available.

What the Master learns automatically

- **Exemplars** — calls scoring ≥85 in post-call QA are auto-promoted into the retrievable knowledge store.
- **Learned transfer** — when a call escalates to a human, the trigger phrase is mined so future paraphrases are caught.
- **Portable brain** — export/import the whole memory as a versioned, idempotent pack to move/stack knowledge between clusters.

10. Reports, transcripts & audit

Report	Use
Agents (`/reports/agents`)	Per-agent productivity, talk/wrap, dispositions.
AI calls (`/reports/ai-calls`)	Every AI-handled call: transcript, QA score, flags, persona, live-switch state.
Recordings (`/reports/recordings`)	Find/play call recordings (retention-governed).
Carrier-race stats	Win rates, prunes — tune carrier racing.
Flagged / TCPA	Compliance review & audit trail.
Search transcripts (`/search/transcripts`)	Full-text search across call transcripts.
Audit (`/audit`)	Every config change & sensitive action, who/when/what.

11. Cluster: nodes & load

Nodes (`/cluster/nodes`) — register each telephony host (capacity, roles). **Load** (`/cluster/load`) — real-time per-node CPU/calls so you balance capacity. Adding a node here is also how you give the AI more horsepower: stand up a stronger box, then point the AI *LLM provider* at it (§9).

12. Settings reference

Setting	What it controls
Telephony	FreeSWITCH/ESL connection, SIP domain, core telephony params.
Pacing	Predictive thresholds & abandon target (global defaults).
Frequency caps	Max attempts per lead/CID per window (compliance).
Wrap-up enforcement	Force/limit after-call work time.
Queue announce	Position/ETA announcements to waiting callers.
Callback	Callback offer & DTMF.
Holidays	No-call dates.
Recording retention	How long recordings are kept (auto-prune).
Carrier-race auto-prune	When to drop a flaky carrier from races.
Database / Backups	DB status; nightly backup config & verify.
SMTP	Outbound email (alerts/reports).
CRM	CRM provider linkage.
Domain / TLS	Public domain & certificate status.
Orgs	Multi-tenant organizations.
Timers	System timing knobs.
Release check	1.0 release-readiness self-check.
AI personas / AI Master	See §9.

13. Operations & maintenance

- **Backups** — verify the nightly backup runs (Settings → Backups → Verify). Keep off-box copies.
- **Recording retention** — set a policy that matches your legal requirement; the sweeper prunes automatically.
- **Database** — monitor size on Settings → Database.
- **Audit** — review periodically; every sensitive change is logged.
- **Updates** — developed and deployed from the VPS; the app rebuilds and the service restarts. No data migration needed for additive changes.
- **AI stack health** — the Master AI readiness preflight is your single source of truth; re-check after any host change.

14. Troubleshooting & the performance/scaling reality

Symptom	Check
AI doesn't pick up calls	Readiness preflight: `mod_audio_stream` loaded? persona enabled+bound? `ai.live_enabled` on? (And remember: AI is overflow — a free human takes the call first.)
AI answers slowly / "thinking"	This is the model on the CPU. Use AI performance (short replies, keep-warm). The real fix is hardware — see below.
AI ignores trained knowledge	Confirm the memory entry is *enabled* and in the right scope (global vs the call's campaign/in-group). Retrieval fires on relevance.
No outbound calls	Carrier reachable? Route plan matches destination? CID group populated? Campaign active & within hours/holidays? Frequency cap not exhausted?
Caller hears nothing	FreeSWITCH/ESL up (Settings → Telephony); DID points at the right in-group/menu.

> **Honest performance note.** Real-time conversational AI is CPU-bound. On a small box (e.g. 5 vCPU), a knowledge-grounded AI turn can take 20–60 s, which is too slow for a live caller — a hardware ceiling, not a bug. The architecture is built for this: stand up a stronger node (a GPU box, or more dedicated cores, running Ollama/vLLM), then on Master AI → **LLM provider** set *Base URL* to that node. One field, no redeploy → correct answers *and* sub-2 s turns. Pick a model that matches the hardware; a smaller model is faster but can answer less accurately, so validate in Mock Call before going live.

15. Quick-start checklists

Go-live: outbound campaign

1. Carrier added & reachable → Route plan → CID group.
2. Create campaign → set pacing, route plan, CID group, dispositions.
3. Import leads; scrub against DNC; set frequency caps & holidays.
4. Assign agents (or bind an AI persona). Activate campaign. Watch Realtime.

Go-live: AI inbound agent

1. Persona created, scripted, voiced, **enabled**; validated in Mock Call.
2. AI-agent user created & bound to the persona; assigned to the in-group serving the DID.
3. Readiness preflight all-green (run `scripts/install-audio-fork.sh` once for `mod_audio_stream`).
4. (Optional) Train the Master via the Training Center; verify in Mock Call.
5. Flip **ai.live_enabled**. Place a test call to the DID with no human available.
